

Finding What Experts Leave Unsaid

Adam Krysztopa · September 2026 · github.com/AdamKrysztopa/edu_proj · Section numbers refer to the Extended Technical Report, which lists the references.

The result

OBS. The absence check ties its mismatched-evidence control; prediction of hidden knowledge is untested. The proof of concept (PoC) built a chain from public web text to ranked questions for experts. Its central step, the absence check (*closure* in the code), asks a judge model whether retrieved ledger sentences already state an element that a lens flagged as missing. In a control, each candidate's additional other-source sentences were swapped for those of the topically nearest candidate, while the seed's own sentences stayed in both arms. The judge's closure rate (counting *partial* as closed) did not change: 0.889, 0.961 and 0.958 in the three ledgers. Under this control, the judge did not tell a candidate's own cross-source evidence from the donor's, so it is not yet an informative absence check; the test does not show why. The control is built into `gapmap/checks.py`, added after an adversarial AI review found the weakness, and it flags the failure on every replay. No gold standard was acquired, so the PoC neither shows nor refutes that the gap map predicts what experts know. A later exploratory spike with a low-cost decision model (Jev) on the committed N3 output gave no per-record signal under its pre-set form test (the score tracked text form). After its first pre-set judge rule failed, it agreed with a second reference, a stronger model, better than the current closure judge did; it detects no hidden knowledge (Section 10.7).

Recommendation: run the closure study before any grant submission

FUT. The cheapest decisive test needs no grant and no participants. In this Phase 0 study (V1), blind coders who are not the owner (two preferred) label each unit as closed, partly closed or open. A unit is one lens firing with its evidence, in one arm of the mismatched-evidence control. The sample is the 107 lens firings from the two admissible ledgers (105 with a donor, in both arms: 210 pools; the firing is the unit of inference), plus about 100 units from one new domain, so that no judge is scored only in-sample. V1 reports three things:

- agreement between the two human coders (weighted Cohen's κ ; the gate uses closed versus not closed);
- corpus-level retrieval recall: is the element stated anywhere in the ledger's sources, not only in the sentences the retriever returned;
- whether any judge in a candidate panel beats the mismatched-evidence control.

The cost is tens of coder hours plus local compute, on the order of a few thousand euros. The gate MS1 reads CONTINUE only if human $\kappa \geq 0.70$, retrieval recall meets its pre-registered floor, and at least one judge beats the mismatched-evidence control by a pre-registered margin. Otherwise the automated absence check stops, and the work pivots to reconstruction with provenance as an audit tool, or to interviewing without a gap map. The V1 numbers should be on page 1 of any grant proposal; this report is the design for them (Section 17).

The problem and the idea

Experts leave out the steps that have become automatic for them. In one small study, three surgeons teaching a procedure omitted on average 73% of the decision steps in a task list built by cognitive task analysis (CTA) (Sullivan et al. 2014); no general omission rate is known. Text has the same silence: writers do not state what is obvious to them (Gordon and Van Durme 2013).

We ask AI for two narrow things. First, *reconstruct* what public sources state about a task, as a ledger of labeled claims whose evidence items are exact spans of saved sources. Second, *predict where that reconstruction is thin*, using constructs from expertise research as methodology lenses. A lens asks: “the sources name this cue, rival cause, rule or check; do they ever state its content?” What survives the absence check becomes a *gap candidate*: a hypothesis labeled *inferred*, with a template question for an expert. What experts would add beyond the ledger is the Human Knowledge Residual, a construct nobody has measured yet.

What the PoC built

OBS. The PoC used zero participants and public web text only, in two domains: intermittent faults in programmable logic controllers (PLC) and data protection impact assessments under the EU General Data Protection Regulation (GDPR), at \$0.50–\$1.00 per run.

Built	Evidence model with computed labels and a frozen schema (N1); reconstruction pipeline with verbatim spans and cross-family verification (N2); gap map with seven lenses, absence check, triage, a breadth ranking (not expected information gain), template questions and built-in checks (N3).
Shown	The chain runs end to end on real sources and replays byte for byte. All 909 stored evidence items re-slice exactly from their snapshots; 260 of 1,169 claims have no locatable span and 158 more have a span the verifier did not accept; all 418 are labeled synthetic. On the two admissible ledgers the lenses fired 107 times and left 25 gap candidates after merging (PLC 24, GDPR v1 1); a third, defect-compromised ledger adds 48 firings and 6 candidates.
Not shown	That any gap candidate is real expert knowledge; that the absence check works; that gating resists planted falsehoods (N2’s gated arms were never scored, so its registered verdict is inconclusive); that the ranking is more than source density; that anything generalizes beyond two domains.

What is new, and what is not. **INT.** Every step has precedent. The closest work checks GDPR privacy policies for typed, regulation-required content (Torre et al. 2020), finds underspecified phrases in instructional text (Anthonio et al. 2022), and mines design rationale from issue trackers (Zhao et al. 2024). What we did not find, in a search to September 2026, is a method that chains a provenance-typed ledger, expertise constructs as absence detectors, a knowledge type and elicitation channel per candidate, and search failures routed back to search. That chain is a proof of concept, not a validated method (Section 4.3).

The program after V1

FUT. Each tranche of money is released only by a gate reading CONTINUE. An external gate reader, an independent statistician, reads the three deciding gates (MS1, MS3, MS5) against their pre-registration; any exception by the owner, at any gate, needs the reader’s sign-off.

- **Phase 1, a lean methods grant** (about EUR 0.3–0.5 million, 12–15 months), only on MS1 CONTINUE. It builds an area score and freezes the configuration before any gold is acquired. It then runs V2: the gap map is scored against several published CTA task lists, pooled, on dated corpora. The gate MS3 only screens money for Phase 2; it does not decide H2, the central hypothesis that the gap map predicts where experts add knowledge. The score is the gain in area under the receiver operating characteristic (ROC) curve (Δ AUROC) over the strongest baseline. MS3 releases Phase 2 only if its point estimate is at least δ (proposed 0.05), its lower 90% bound is above $-\delta/2$, and the map adds value beyond source density and area size. Otherwise H2 funding stops and the separately costed interview-only pivot follows.

- **Phase 2, the full program**, only on MS3 CONTINUE. Blind prospective elicitation (V5), with a provisional smallest gain of interest $\delta = 0.05$ and a size set after its pilot, is the prospective test of H2. H2 stops if the upper 90% bound of ΔAUROC is below δ ; any reading short of CONTINUE does not support H2 and ends its funding. A later study tests H3: prioritized questions recover at least 1.25 times as many validated items per expert hour; a null counts against it. Organizational and education pilots follow, each with its own gate.

A study of developer departures in open-source projects tests a separate organizational hypothesis (H-OSS) and can neither stop nor license H2. Budgets: Section 25. **Team status:** one operator built the PoC with AI assistance; a host institution, principal investigator, statistician, CTA collaborator and letters must be named before submission (Section 24).